MCC Interim Linux / Linux 1.0.4 Complete Modern Installation Guide

This document contains the FULL process used to successfully install and boot MCC Interim Linux with Linux kernel 1.0.4 on modern hardware using the Bochs emulator. The installation process included:

• Booting from floppy images
• Debugging failed console initialization
• Solving floppy swapping issues
• Creating a virtual hard disk
• Partitioning the drive manually
• Creating an ext2 filesystem
• Installing Linux binaries
• Installing and configuring LILO
• Successfully booting Linux 1.0.4 from a hard disk image

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1. Why This Project Exists

MCC Interim Linux is one of the earliest Linux distributions ever created. Very little modern documentation exists for installing it today. Most information available online is:

• Extremely outdated
• Stored on old mirrors
• Missing important installation details
• Written for real 1990s hardware instead of modern PCs

The goal of this guide is to preserve a FULL working installation process that modern users can reproduce.

2. Why QEMU Failed

The installation was originally attempted with QEMU. Problems encountered:

• Boot freezing at "LI"
• Kernel hangs during initialization
• Console initialization failures
• Unstable floppy handling

QEMU is optimized for newer operating systems and modern virtualization. Very early Linux releases sometimes expect older BIOS behavior and older hardware timing. Bochs worked significantly better because:

• It emulates older hardware more accurately
• It handles legacy BIOS behavior better
• It is slower but historically more compatible

3. Software Used

Required software:

• Bochs 3.0
• qemu-img utility
• MCC Interim Linux floppy images
• Windows system for hosting the emulator

4. Creating the Virtual Hard Disk

The hard disk image was created using qemu-img. A raw disk image was chosen because it works reliably with Bochs.

```
qemu-img create -f raw "C:\Users\amine\Downloads\mcc-hdd.img" 20M
```

5. Configuring Bochs

The hard disk image must be added manually to the Bochs configuration file. Without this step, Linux will not detect the hard drive correctly.

```
ata0-master: type=disk, path="C:\Users\amine\Downloads\mcc-hdd.img", mode=flat, cylinders=306, heads=4, spt=17
```

6. First Boot Attempt

The system was booted using the MCC Interim Linux floppy image. At first:

• The screen occasionally turned completely black
• Booting was extremely slow
• The installer sometimes froze during initialization

Eventually the system reached the Linux boot messages successfully.

7. Console Initialization Errors

One of the biggest problems encountered was:

"Unable to open an initial console."

Several boot parameters were tested:

• ramdisk console=ttys0
• ramdisk console=tty0

These did not fully solve the issue. The real fix came from using the correct floppy images and reboot sequence.

8. Floppy Swapping Process

The installer required multiple floppy disk swaps. Common mistakes included:

• Ejecting the wrong floppy drive
• Forgetting to insert the root disk
• Pressing ENTER before the correct disk was inserted

At one point the installer repeatedly produced:

• MINIX filesystem errors
• "Unable to read superblock"
• "Mount failed"

The issue was solved by correctly swapping floppy B instead of floppy A.

9. Entering the MCC Installer

After the floppy swap issue was fixed, the MCC Interim Linux installer menu finally appeared. The installer provided options such as:

• Run fdisk
• Set up swap
• Create filesystem
• Install binaries
• Run shell scripts

10. Partitioning the Disk

fdisk was used to manually create the Linux partition. Process:

1. Create a new primary partition
2. Use the entire virtual disk
3. Set Linux partition type
4. Write the partition table

Several warnings appeared during this process:

• Invalid partition table flags
• Cylinder alignment warnings

These warnings are normal for extremely old Linux tools.

11. Creating the ext2 Filesystem

The filesystem was created manually on /dev/hda1. During creation:

• inode tables were written
• superblocks were generated
• badblocks checks were performed

The installer then mounted the new filesystem successfully.

12. Installing Linux Binaries

The MCC installer copied the Linux binaries from floppy disks to the hard drive. During installation:

• base.tgz packages were extracted
• filesystem checks were performed
• keyboard configuration was requested

The installer occasionally displayed keyboard scan code warnings, but these did not prevent installation.

13. Installing LILO

LILO was installed directly to /dev/hda. This was one of the most important steps because:

• Without LILO the system would not boot from the hard disk
• The floppy disks would always be required

Linux boot entries were configured during the installation process.

14. First Successful Boot

After rebooting:

• The floppy disks were removed
• Bochs booted directly from the virtual hard disk
• Linux 1.0.4 started successfully

The login prompt finally appeared:

linux login:

15. Logging Into Linux

The system was accessed using:

username: root

Once logged in:

• uname -a confirmed Linux 1.0.4
• ls / displayed the filesystem structure
• mount displayed mounted filesystems
• dmesg displayed kernel boot logs

16. Important Commands Used

Important commands used during testing:

Command	Purpose
uname -a	Display kernel version information
ls /	List root filesystem directories
mount	Show mounted filesystems
dmesg	Display kernel boot messages
fdisk	Partition the virtual hard disk

17. GitHub Preservation Project

After the successful installation, the entire project was preserved publicly on GitHub. The repository includes:

• Working HDD image
• Original floppy images
• Bochs configuration
• Screenshots
• Documentation
• Troubleshooting notes

Repository: github.com/aminewe898/mcc-interim-linux-modern-guide

18. Why This Matters

This project preserves a small but important part of Linux history. Modern Linux users rarely experience:

• Floppy disk booting
• Manual partitioning from scratch
• Primitive bootloaders
• Early kernel behavior

This installation process provides a closer look at what Linux users actually experienced in 1994.

19. Final Thoughts

What started as a small retrocomputing experiment became a complete preservation project. The final result:

• Fully working Linux 1.0.4 installation
• Bootable hard disk image
• Public GitHub repository
• Modern installation documentation

The project successfully demonstrates that extremely old Linux systems can still be installed and explored today using modern hardware and emulators.